

six fan driver board

Arduino Nano carrier: 6 low-side MOSFET fan channels, 12 V fans, 5 V logic from the supply

POWER INPUT

J1 PWR 12V/5V
1 +12V 2 GND 3 +5V 4 GND

C1 100uF 25V C2 10uF 25V C3 100nF C4 10uF 25V C5 100nF

PWR_FLAG +12V GND

ARDUINO NANO (plugs into J2/J3)

A1 A7 D0/RX D1/TX D2 D3 D4 D5 D6 D7 D8 D9 D10 D11 D12 D13 RESET RESET AREF AO AO A1 A2 A3 A4 A5 A6 A7 VIN 3V3 5V GND GND

FAN1_PWM FAN2_PWM FAN3_PWM FAN4_PWM FAN5_PWM FAN6_PWM

J2 Socket 1x15
J3 Socket 1x15

MOUNTING

H1 H2 M3 M3 H3 H4 M3 M3

FAN CHANNELS: J11-J16 pin 1 = +12V, pin 2 = switched return

CHANNEL 1 (Nano D3, connector J11)

FAN1_PWM R1 1k GATE1 GND R11 100k Q1 AO3400A FAN1_DRV SS14 D11 Red LED1_A R21 4.7k J11 B2B-XH-A

CHANNEL 2 (Nano D5, connector J12)

FAN2_PWM R2 1k GATE2 GND R12 100k Q2 AO3400A FAN2_DRV SS14 D12 Red LED2_A R22 4.7k J12 B2B-XH-A

CHANNEL 3 (Nano D6, connector J13)

FAN3_PWM R3 1k GATE3 GND R13 100k Q3 AO3400A FAN3_DRV SS14 D13 Red LED3_A R23 4.7k J13 B2B-XH-A

CHANNEL 4 (Nano D9, connector J14)

FAN4_PWM R4 1k GATE4 GND R14 100k Q4 AO3400A FAN4_DRV SS14 D14 Red LED4_A R24 4.7k J14 B2B-XH-A

CHANNEL 5 (Nano D10, connector J15)

FAN5_PWM R5 1k GATE5 GND R15 100k Q5 AO3400A FAN5_DRV SS14 D15 Red LED5_A R25 4.7k J15 B2B-XH-A

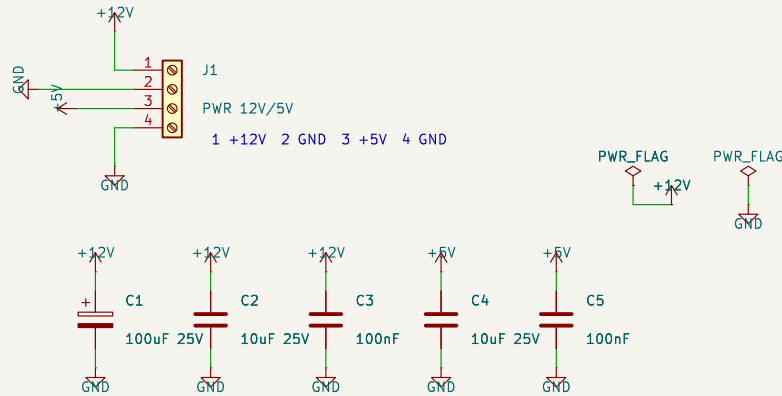
CHANNEL 6 (Nano D11, connector J16)

FAN6_PWM R6 1k GATE6 GND R16 100k Q6 AO3400A FAN6_DRV SS14 D16 Red LED6_A R26 4.7k J16 B2B-XH-A

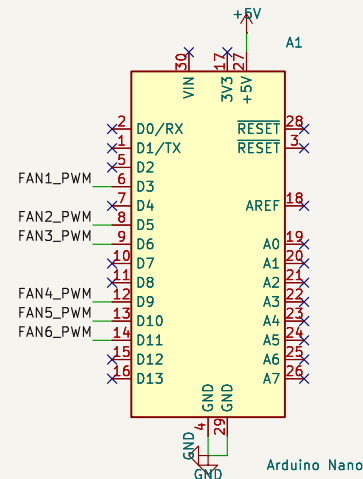
Source: <https://github.com/ideocentric/four-tone>
Parts: JLCPCB/LCSC numbers in the LCSC field
CERN-OHL-S-2.0
four-tone project
Sheet: /
File: four-tone-driver.kicad_sch
Title: six fan driver
Size: A3 Date: 2026-09-15 Rev: A
KiCad E.D.A. 9.0.6 Id: 1/1

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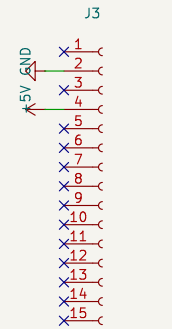
POWER INPUT



ARDUINO NANO (plugs into J2/J3)



Socket 1x15

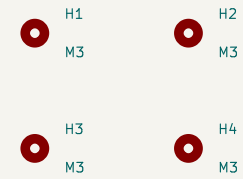


Socket 1x15

J2 pin k = Nano pin k. J3 pin k = Nano pin 31-k.

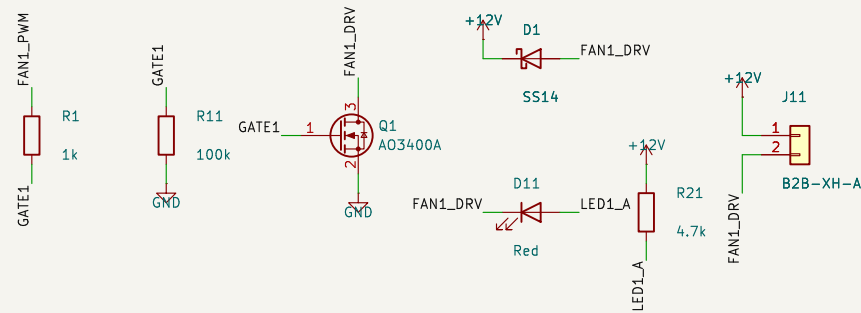
Place J2/J3 pin 1 on the same row, 15.24 mm apart (Module:Arduino_Nano pads 1 and 30).

MOUNTING

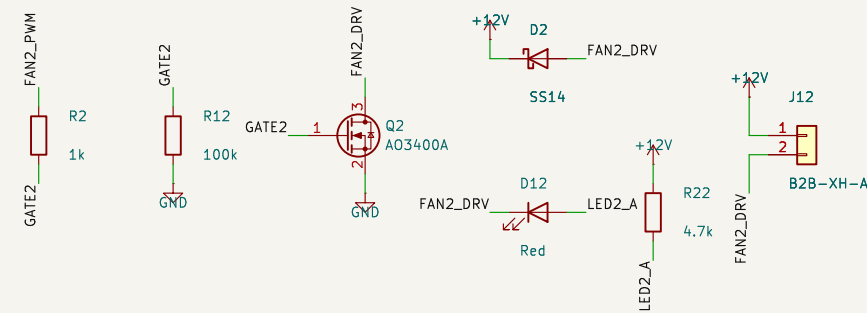


FAN CHANNELS: J11–J16 pin 1 = +12V, pin 2 = switched return

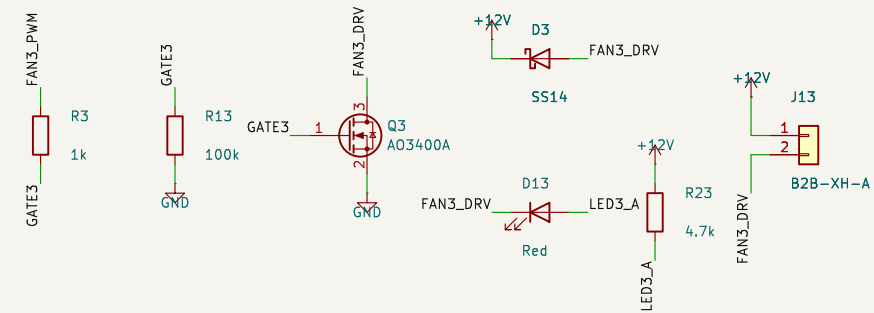
CHANNEL 1 (Nano D3, connector J11)



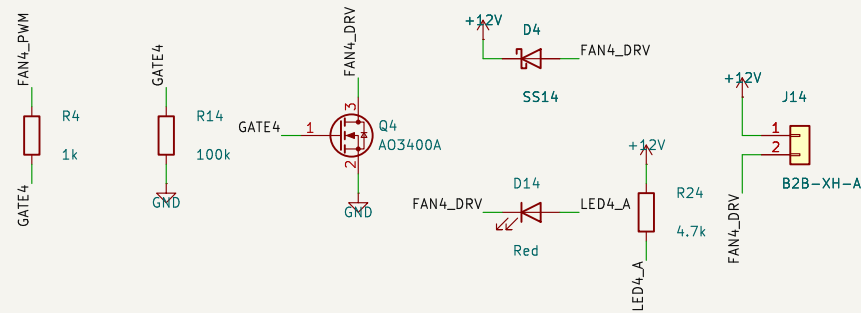
CHANNEL 2 (Nano D5, connector J12)



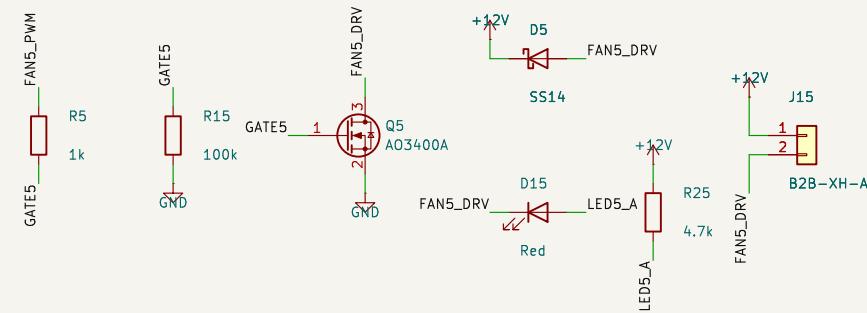
CHANNEL 3 (Nano D6, connector J13)



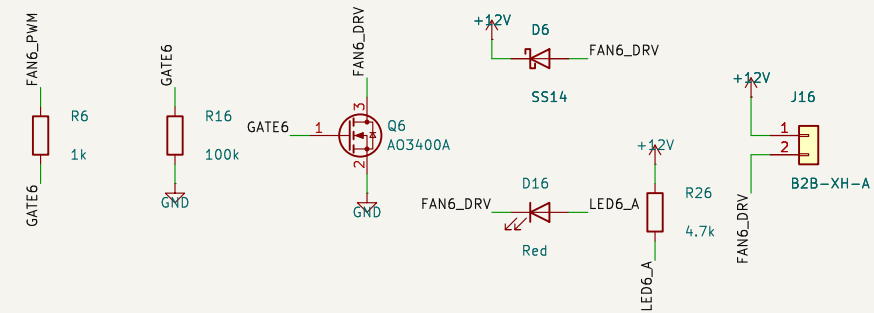
CHANNEL 4 (Nano D9, connector J14)



CHANNEL 5 (Nano D10, connector J15)



CHANNEL 6 (Nano D11, connector J16)



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KiCad E.D.A. 9.0.6

Rev: A

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